

DEFINITION 1

A *set* is an unordered collection of objects, called *elements* or *members* of the set. A set is said to *contain* its elements. We write $a \in A$ to denote that a is an element of the set A . The notation $a \notin A$ denotes that a is not an element of the set A .

The Size of a Set

Sets are used extensively in counting problems, and for such applications we need to discuss the sizes of sets.

DEFINITION 4

Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a *finite set* and that n is the *cardinality* of S . The cardinality of S is denoted by $|S|$.

Remark: The term *cardinality* comes from the common usage of the term *cardinal number* as the size of a finite set.

EXAMPLE 10 Let A be the set of odd positive integers less than 10. Then $|A| = 5$. 

EXAMPLE 11 Let S be the set of letters in the English alphabet. Then $|S| = 26$. 

EXAMPLE 12 Because the null set has no elements, it follows that $|\emptyset| = 0$. 

We will also be interested in sets that are not finite.

DEFINITION 5

A set is said to be *infinite* if it is not finite.

EXAMPLE 13 The set of positive integers is infinite. 



We will extend the notion of cardinality to infinite sets in Section 2.5, a challenging topic full of surprising results.

Power Sets

Many problems involve testing all combinations of elements of a set to see if they satisfy some property. To consider all such combinations of elements of a set S , we build a new set that has as its members all the subsets of S .

DEFINITION 6

Given a set S , the *power set* of S is the set of all subsets of the set S . The power set of S is denoted by $\mathcal{P}(S)$.

EXAMPLE 14 What is the power set of the set $\{0, 1, 2\}$?



Solution: The power set $\mathcal{P}(\{0, 1, 2\})$ is the set of all subsets of $\{0, 1, 2\}$. Hence,

$$\mathcal{P}(\{0, 1, 2\}) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}.$$

Note that the empty set and the set itself are members of this set of subsets. 

EXAMPLE 15 What is the power set of the empty set? What is the power set of the set $\{\emptyset\}$?

Solution: The empty set has exactly one subset, namely, itself. Consequently,

$$\mathcal{P}(\emptyset) = \{\emptyset\}.$$

The set $\{\emptyset\}$ has exactly two subsets, namely, $\emptyset$ and the set $\{\emptyset\}$ itself. Therefore,

$$\mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}. \quad \text{img alt="arrow" data-bbox="738 824 751 834" style="float: right; margin-left: 10px;}" data-bbox="738 824 751 834"/>$$

If a set has n elements, then its power set has 2^n elements. We will demonstrate this fact in several ways in subsequent sections of the text.

DEFINITION 8

Let A and B be sets. The *Cartesian product* of A and B , denoted by $A \times B$, is the set of all ordered pairs (a, b) , where $a \in A$ and $b \in B$. Hence,

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}.$$

EXAMPLE 16

Let A represent the set of all students at a university, and let B represent the set of all courses offered at the university. What is the Cartesian product $A \times B$ and how can it be used?



Solution: The Cartesian product $A \times B$ consists of all the ordered pairs of the form (a, b) , where a is a student at the university and b is a course offered at the university. One way to use the set $A \times B$ is to represent all possible enrollments of students in courses at the university. ◀

EXAMPLE 17

What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b, c\}$?

Solution: The Cartesian product $A \times B$ is

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

Note that the Cartesian products $A \times B$ and $B \times A$ are not equal, unless $A = \emptyset$ or $B = \emptyset$ (so that $A \times B = \emptyset$) or $A = B$ (see Exercises 31 and 38). This is illustrated in Example 18.

EXAMPLE 18

Show that the Cartesian product $B \times A$ is not equal to the Cartesian product $A \times B$, where A and B are as in Example 17.

Solution: The Cartesian product $B \times A$ is

$$B \times A = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}.$$

This is not equal to $A \times B$, which was found in Example 17. ◀

EXAMPLE 19

What is the Cartesian product $A \times B \times C$, where $A = \{0, 1\}$, $B = \{1, 2\}$, and $C = \{0, 1, 2\}$?

Solution: The Cartesian product $A \times B \times C$ consists of all ordered triples (a, b, c) , where $a \in A$, $b \in B$, and $c \in C$. Hence,

$$A \times B \times C = \{(0, 1, 0), (0, 1, 1), (0, 1, 2), (0, 2, 0), (0, 2, 1), (0, 2, 2), (1, 1, 0), (1, 1, 1), (1, 1, 2), (1, 2, 0), (1, 2, 1), (1, 2, 2)\}.$$

DEFINITION 1

Let A and B be sets. The **union** of the sets A and B , denoted by $A \cup B$, is the set that contains those elements that are either in A or in B , or in both.

An element x belongs to the union of the sets A and B if and only if x belongs to A or x belongs to B . This tells us that

$$A \cup B = \{x \mid x \in A \vee x \in B\}.$$

The Venn diagram shown in Figure 1 represents the union of two sets A and B . The area that represents $A \cup B$ is the shaded area within either the circle representing A or the circle representing B .

We will give some examples of the union of sets.

EXAMPLE 1

The union of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{1, 2, 3, 5\}$; that is, $\{1, 3, 5\} \cup \{1, 2, 3\} = \{1, 2, 3, 5\}$. ◀

EXAMPLE 2

The union of the set of all computer science majors at your school and the set of all mathematics majors at your school is the set of students at your school who are majoring either in mathematics or in computer science (or in both). ▶

DEFINITION 2

Let A and B be sets. The **intersection** of the sets A and B , denoted by $A \cap B$, is the set containing those elements in both A and B .

An element x belongs to the intersection of the sets A and B if and only if x belongs to A and x belongs to B . This tells us that

$$A \cap B = \{x \mid x \in A \wedge x \in B\}.$$

EXAMPLE 15

Let $A = \{0, 2, 4, 6, 8\}$, $B = \{0, 1, 2, 3, 4\}$, and $C = \{0, 3, 6, 9\}$. What are $A \cup B \cup C$ and $A \cap B \cap C$?

Solution: The set $A \cup B \cup C$ contains those elements in at least one of A , B , and C . Hence,

$$A \cup B \cup C = \{0, 1, 2, 3, 4, 6, 8, 9\}.$$

The set $A \cap B \cap C$ contains those elements in all three of A , B , and C . Thus,

$$A \cap B \cap C = \{0\}. \quad \blacktriangleleft$$

DEFINITION 1

Let A and B be nonempty sets. A **function** f from A to B is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A . If f is a function from A to B , we write $f : A \rightarrow B$.

DEFINITION 2

If f is a function from A to B , we say that A is the **domain** of f and B is the **codomain** of f . If $f(a) = b$, we say that b is the **image** of a and a is a **preimage** of b . The **range**, or **image**, of f is the set of all images of elements of A . Also, if f is a function from A to B , we say that f **maps** A to B .

DEFINITION 3 Let f_1 and f_2 be functions from A to $\mathbf{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbf{R}$ defined for all $x \in A$ by

$$\begin{aligned}(f_1 + f_2)(x) &= f_1(x) + f_2(x), \\ (f_1 f_2)(x) &= f_1(x) f_2(x).\end{aligned}$$

Note that the functions $f_1 + f_2$ and $f_1 f_2$ have been defined by specifying their values at x in terms of the values of f_1 and f_2 at x .

EXAMPLE 6 Let f_1 and f_2 be functions from $\mathbf{R}$ to $\mathbf{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?

Solution: From the definition of the sum and product of functions, it follows that

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$$

and

$$(f_1 f_2)(x) = x^2(x - x^2) = x^3 - x^4.$$

When f is a function from A to B , the image of a subset of A can also be defined.

DEFINITION 4 Let f be a function from A to B and let S be a subset of A . The *image* of S under the function f is the subset of B that consists of the images of the elements of S . We denote the image of S by $f(S)$, so

$$f(S) = \{t \mid \exists s \in S (t = f(s))\}.$$

We also use the shorthand $\{f(s) \mid s \in S\}$ to denote this set.

Remark: The notation $f(S)$ for the image of the set S under the function f is potentially ambiguous. Here, $f(S)$ denotes a set, and not the value of the function f for the set S .

EXAMPLE 7 Let $A = \{a, b, c, d, e\}$ and $B = \{1, 2, 3, 4\}$ with $f(a) = 2$, $f(b) = 1$, $f(c) = 4$, $f(d) = 1$, and $f(e) = 1$. The image of the subset $S = \{b, c, d\}$ is the set $f(S) = \{1, 4\}$.

One-to-One and Onto Functions

Some functions never assign the same value to two different domain elements. These functions are said to be **one-to-one**.

DEFINITION 5 A function f is said to be *one-to-one*, or an *injection*, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be *injective* if it is one-to-one.

Inverse Functions and Compositions of Functions

Now consider a one-to-one correspondence f from the set A to the set B . Because f is an onto function, every element of B is the image of some element in A . Furthermore, because f is also a one-to-one function, every element of B is the image of a *unique* element of A . Consequently, we can define a new function from B to A that reverses the correspondence given by f . This leads to Definition 9.

DEFINITION 9

Let f be a one-to-one correspondence from the set A to the set B . The *inverse function* of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$.

Remark: Be sure not to confuse the function f^{-1} with the function $1/f$, which is the function that assigns to each x in the domain the value $1/f(x)$. Notice that the latter makes sense only when $f(x)$ is a non-zero real number.

Figure 6 illustrates the concept of an inverse function.

If a function f is not a one-to-one correspondence, we cannot define an inverse function of f . When f is not a one-to-one correspondence, either it is not one-to-one or it is not onto. If f is not one-to-one, some element b in the codomain is the image of more than one element in the domain. If f is not onto, for some element b in the codomain, no element a in the domain exists for which $f(a) = b$. Consequently, if f is not a one-to-one correspondence, we cannot assign to each element b in the codomain a unique element a in the domain such that $f(a) = b$ (because for some b there is either more than one such a or no such a).

A one-to-one correspondence is called **invertible** because we can define an inverse of this function. A function is **not invertible** if it is not a one-to-one correspondence, because the inverse of such a function does not exist.

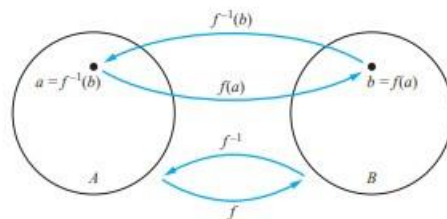


FIGURE 6 The Function f^{-1} Is the Inverse of Function f .

EXAMPLE 18 Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible, and if it is, what is its inverse?

Solution: The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence given by f , so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$. ◀

EXAMPLE 19 Let $f : \mathbf{Z} \rightarrow \mathbf{Z}$ be such that $f(x) = x + 1$. Is f invertible, and if it is, what is its inverse?

Solution: The function f has an inverse because it is a one-to-one correspondence, as follows from Examples 10 and 14. To reverse the correspondence, suppose that y is the image of x , so that $y = x + 1$. Then $x = y - 1$. This means that $y - 1$ is the unique element of $\mathbf{Z}$ that is sent to y by f . Consequently, $f^{-1}(y) = y - 1$. ◀

EXAMPLE 20 Let f be the function from $\mathbf{R}$ to $\mathbf{R}$ with $f(x) = x^2$. Is f invertible?

Solution: Because $f(-2) = f(2) = 4$, f is not one-to-one. If an inverse function were defined, it would have to assign two elements to 4. Hence, f is not invertible. (Note we can also show that f is not invertible because it is not onto.) ◀

DEFINITION 10

Let g be a function from the set A to the set B and let f be a function from the set B to the set C . The *composition* of the functions f and g , denoted for all $a \in A$ by $f \circ g$, is defined by

$$(f \circ g)(a) = f(g(a)).$$

FIGURE 1 The Composition of the Functions f and g .

EXAMPLE 22

Let g be the function from the set $\{a, b, c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a, b, c\}$ to the set $\{1, 2, 3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$. What is the composition of f and g , and what is the composition of g and f ?

Solution: The composition $f \circ g$ is defined by $(f \circ g)(a) = f(g(a)) = f(b) = 2$, $(f \circ g)(b) = f(g(b)) = f(c) = 1$, and $(f \circ g)(c) = f(g(c)) = f(a) = 3$.

Note that $g \circ f$ is not defined, because the range of f is not a subset of the domain of g . ◀

EXAMPLE 23

Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? What is the composition of g and f ?

Solution: Both the compositions $f \circ g$ and $g \circ f$ are defined. Moreover,

$$(f \circ g)(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

and

$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11. \quad \blacktriangleleft$$

Remark: Note that even though $f \circ g$ and $g \circ f$ are defined for the functions f and g in Example 23, $f \circ g$ and $g \circ f$ are not equal. In other words, the commutative law does not hold for the composition of functions.

1.2 SETS

A set is any *well-defined* collection of objects, known as *elements* or *members* of the set. We generally use capital letters, A, B, X, Y , to denote sets and lowercase letters, $a, b, x, y, \dots$, to denote elements of sets. Here, the term *well-defined* implies that there exists certain rule by which it can be decided that whether a given object or element belongs to the collection or not. The statement ' x is an element of A ' or, equivalently, ' x belongs to A ' is written as $x \in A$. The statement ' x is not an element of A ' is written as $x \notin A$.

1.2.1 Representation of a Set

There are mainly two types of representation of a set:

- I. Roster or tabular form
- II. Set builder form or rule method

I. *Roster or tabular form* All the members of the set, here, are listed and being separated by commas, and are enclosed within braces.

For example The notation $\{a, b, c, d\}$ represents the set with the four elements a, b, c , and d . Moreover,

- (i) The set V of all vowels in the English alphabet can be written as $V = \{a, e, i, o, u\}$.
- (ii) The set O of odd positive integers less than 8 can be represented by $O = \{1, 3, 5, 7\}$.

II. *Set builder or rule method* In this method, we define certain property which recognises the elements in the set.

For example

- (i) Let $B = \{x : x \text{ is an even integer, } x > 0\}$, which implies that ' B is the set of x such that x is an even integer and x is greater than 0'. Here, colon ($:$) is read as 'such that' and comma ($,$) as 'and'. A vertical bar is also used in place of colon.
- (ii) The set $X = \{x : x \text{ is a real number, } 1 < x < 8\}$ is the set of real numbers that lie strictly between 1 and 8.



EXAMPLE 1.1 Redefine each of the following sets using set-builder notation:

- (i) $\{2, 3, 4, 6, 8, 9, 10, \dots\}$
- (ii) $\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots\right\}$
- (iii) $\{0, 1, 2, \dots, 50\}$

Solution

- (i) $\{x : x = 2n \text{ or } x = 3n \text{ for } n \in \mathbb{N}\}$
- (ii) $\left\{x : x = \frac{1}{n} \text{ for } n \in \mathbb{N}\right\}$
- (iii) $\{x : 0 \leq x \leq 50 \text{ and } x \in \mathbb{Z}\}$

1.2.2 Sets of Special Status

There are some important sets which can be represented by special name:

$\mathbb{Z}$, the set of all integers = $\{\dots, -2, -1, 0, 1, 2, \dots\}$

$\mathbb{Z}^+$, the set of positive integers = $\{1, 2, 3, 4, \dots\}$
 $\mathbb{N}$, the set of natural numbers = $\{0, 1, 2, 3, 4, \dots\}$
 $\mathbb{R}$, the set of real numbers
 $\mathbb{Q}$, the set of rational numbers = $\{p/q : p \in \mathbb{Z}, q \in \mathbb{Z}, \text{ and } q \neq 0\}$
 $\mathbb{C}$, the set of complex numbers

1.2.3 Universal Set and Empty Set

If there are some sets under consideration, then there happens to be a fixed set which contains each one of the given sets. Such a fixed set is known as the *universal set*, and is denoted by U or ξ .

For example

- (i) In human population studies, the universal set consists of all the people in the world.
 (ii) If $A = \{1, 2, 3, 4\}$, $B = \{2, 3, 5, 7\}$, and $C = \{2, 4, 6, 8\}$, then $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ is the universal set.

A set consisting of no element at all is called an *empty set* or a *null set* or a *void set* and is denoted by ϕ . In roster form, ϕ is denoted by $\{\}$.

For example

- (i) $\{x : x \in \mathbb{N}, 2 < x < 3\} = \phi$
 (ii) $\{x : x \in \mathbb{R}, x^2 = -1\} = \phi$

A set which possesses at least one element is called a *non-empty set*. A set consisting of a single element is called a *singleton set*.

For example The set $\{0\}$ is a singleton set, whose only member is 0.

1.2.4 Subsets

If A and B are two sets such that every element in a set A is also an element of a set B , then A is called a *subset* of B . We also say that A is contained in B or that B contains A . This relationship is expressed as

$$A \subseteq B \text{ or } B \supseteq A$$

If there exists even a single element in A , which is not in B , then A is not in B ; hence A is not a subset of B and we write

$$A \not\subseteq B \text{ or } B \not\supseteq A$$

If $A \subseteq B$, then it is possible that $A = B$. When $A \subseteq B$, but $A \neq B$, we say that A is a *proper subset* of B . Again, if $A \subset B$, then A is a proper subset of B .

For example Let $A = \{1, 3\}$, $B = \{1, 2, 3\}$, and $C = \{1, 3, 2\}$. Then A and B are both subsets of C , but A is a proper subset of C . On the other hand, since $B = C$, we cannot say that B is a proper subset of C .

The following properties of set are very important:

- I. Every set is a subset of itself.
- II. The empty set is a subset of every set.
- III. The total number of all possible subsets of a given set containing n elements is 2^n .

Proof (Property III) Let A be any set containing n elements. Then, one of its subsets is the empty set. Apart from this,

the number of singleton subsets of $A = n = {}^nC_1$,

the number of subsets of A , each containing 2 elements $= {}^nC_2$,

the number of subsets of A , each containing 3 elements $= {}^nC_3$,

...

the number of subsets of A , each containing n elements $= {}^nC_n$.

Hence, the total number of all possible subsets of A

$$= (1 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n)$$

$$= (1 + 1)^n = 2^n$$

using binomial theorem.

For example If the set $A = \{1, 2, 3, 4\}$ contains 4 elements, then the total number of its subsets is $2^4 = 16$.

Note A set A with n elements has 2^n subsets.

1.2.5 Power Set

Consider the set $A = \{a, b\}$. The subsets of A are ϕ , $\{a\}$, $\{b\}$, and $\{a, b\}$. Then, the family of all the subsets of A is called the *power set* of A , which is denoted by $P(A)$. Thus, $P(A) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$. Symbolically, $P(A) = \{x : x \text{ is a subset of } A\}$.

It may be noted here that both the empty set and the set A themselves are members of the power set.

For example Consider $A = \phi$. Then $P(A) = \{\phi\}$ and contains one element, the empty set itself.

EXAMPLE 1.2 List all the members of the power set of each of the following sets:
(i) $A = \{1, 2, 3\}$ (ii) $B = \{\{a\}, \{b\}\}$ (iii) $C = \{\phi\}$

Solution

(i) $P(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, A\}$

(ii) $P(B) = \{\phi, \{a\}, \{b\}, B\}$

(iii) $P(C) = \{\phi, C\}$

Note If the set A is finite and contains n elements, then the power set of A will have 2^n elements.

1.2.6 Cardinality of a Set

The number of distinct elements contained in a finite set is called the *cardinality* or the *cardinal number* of the set. The cardinality of a set is denoted by various notations like, $n(A)$ or, $\text{card}(A)$, $|A|$, and A .

For example Cardinality of empty set, ϕ , is 0 and is denoted by $n(\phi) = 0$. Let $A = \{2, 3, 5, 7, 11\}$, then $n(A) = 5$.

1.3 ORDERED PAIRS

An *ordered pair* of objects is a pair of objects arranged in some order. Thus, in the set $\{a, b\}$ of two objects, a is the first one and b is the second object of a pair. Hence, (a, b) and (b, a) are two different ordered pairs. Moreover, the two objects in ordered pair need not be distinct. So, (a, a) is a well-defined ordered pair.

An ordered triple is an ordered triple of objects $\{a, b, c\}$ where a is the first object, b is the second one, and c is the third element of triple. An ordered triple can also be written in terms of ordered pair as $\{(a, b), c\}$. Similarly, an ordered quadruple is an ordered pair $\{((a, b), c), d\}$ with first element as ordered triple. Thus, an ordered n -tuple is an ordered pair whose first component is an ordered $(n - 1)$ -tuple.

Let us now define an *ordered set* and simultaneously redefine an ordered pair. Actually, an ordered set is defined as an ordered collection of distinct objects. In general, an ordered n -tuple is an ordered set with n elements. First component of this n -tuple is an ordered $(n - 1)$ -tuple and the n th element is the second component. Also, an ordered set of n elements is an ordered pair of $(n - 1)$ -tuple and an element.

For example An ordered set of 5 elements $\{a, b, c, d, e\}$ can be expressed as $\{(((a, b), c), d), e\}$.

Many of the discrete structures, we will study in later chapters, are based on the notion of the *cartesian product of sets*.

1.3.1 Cartesian Product of Sets

Cartesian product is a mode by which two or more sets can be combined to obtain another one. If A and B are two non-empty sets, the *cartesian product* (or cross product or direct product) of A and B is the set,

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$$

If $A = \phi$ or $B = \phi$, we define $A \times B = \phi$.

Note

- (i) If A and B are finite sets, then $n(A \times B) = n(A) \cdot n(B)$.
- (ii) If either A or B is an infinite set, then $A \times B$ is an infinite set.

The cartesian product of $A \times A$ is denoted by A_2 . More generally,

$$A_n = A \times A \times \cdots \times A (n \text{ times}) = \{(a_1, a_2, \dots, a_n) : a_i \in A, i = 1, 2, \dots, n\}$$

Let us redefine the cartesian product in terms of ordered pairs.

The cartesian product of A and B is denoted by $A \times B$ and is the set of all ordered pairs of the form (a, b) , where $a \in A$ and $b \in B$.

For example Let $A = \{a, b\}$ and $B = \{a, c, d\}$. Then,

$$A \times B = \{a, b\} \times \{a, c, d\} = \{(a, a), (a, c), (a, d), (b, a), (b, c), (b, d)\}$$

and

$$B \times A = \{a, c, d\} \times \{a, b\} = \{(a, a), (a, b), (c, a), (c, b), (d, a), (d, b)\}$$

The elements of $A \times B$ are ordered pairs of the order count, i.e., $(a, b) \neq (b, a)$ unless $a = b$.
 For example A software firm used to provide the following three characteristics for each program that it sold:

Language FORTRAN (f); PASCAL (p); LISP (l)

Memory 2 meg (2); 4 meg (4); 8 meg (8)

Operating system UNIX (u); DOS (d)

Let $L = \{f, p, l\}$, $M = \{2, 4, 8\}$, and $O = \{u, d\}$. Then the cartesian product $L \times M \times O$ provides all the categories that describe a program. There are $3 \cdot 3 \cdot 2$ or 18 categories in this classification scheme.



EXAMPLE 1.3 If $A = \{1, 2, 3\}$, $B = \{3, 4\}$, and $C = \{4, 5, 6\}$, determine

- (i) $A \times (B \cup C)$ (ii) $A \times (B \cap C)$ (iii) $(A \times B) \cap (B \times C)$

Solution

$$(i) B \cup C = \{3, 4\} \cup \{4, 5, 6\} = \{3, 4, 5, 6\}$$

Thus,

$$\begin{aligned} A \times (B \cup C) &= \{1, 2, 3\} \times \{3, 4, 5, 6\} \\ &= \{(1, 3), (1, 4), (1, 5), (1, 6), (2, 3), (2, 4), (2, 5), (2, 6), (3, 3), (3, 4), (3, 5), (3, 6)\} \end{aligned}$$

$$(ii) B \cap C = \{3, 4\} \cap \{4, 5, 6\} = \{4\}$$

Thus,

$$A \times (B \cap C) = \{1, 2, 3\} \times \{4\} = \{(1, 4), (2, 4), (3, 4)\}$$

$$(iii) A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4), (3, 3), (3, 4)\}$$

and

$$B \times C = \{(3, 4), (3, 5), (3, 6), (4, 4), (4, 5), (4, 6)\}$$

Thus,

$$(A \times B) \cap (B \times C) = \{3, 4\}$$

1.3.2 Properties of Cartesian Product

I. For any three sets A, B, C

- (i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$
 (ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

Proof

(i) Let a and b be arbitrary elements of $A \times (B \cup C)$. Then,

$$(a, b) \in A \times (B \cup C)$$

$$\Rightarrow a \in A \text{ and } b \in (B \cup C)$$

$$\Rightarrow a \in A \text{ and } (b \in B \text{ or } b \in C)$$

$$\Rightarrow (a \in A \text{ and } b \in B) \text{ or } (a \in A \text{ and } b \in C)$$

$$\Rightarrow (a, b) \in (A \times B) \text{ or } (a, b) \in (A \times C)$$

$$\Rightarrow (a, b) \in (A \times B) \cup (A \times C)$$

Thus,

$$A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)$$

Similarly,

$$(A \times B) \cup (A \times C) \subseteq A \times (B \cup C)$$

Hence,

$$A \times (B \cup C) = (A \times B) \cup (A \times C)$$

(ii) This may be proved in a similar manner and is left to the reader.

II. For any three sets A, B, C , $A \times (B - C) = (A \times B) - (A \times C)$. The proof is left to the reader.

III. For any sets A, B, C, D , $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.

Proof Let $(a, b) \in (A \times B) \cap (C \times D)$, i.e., $(a, b) \in (A \times B)$ and $(a, b) \in (C \times D)$

$$\Rightarrow (a \in A \text{ and } b \in B) \text{ and } (a \in C \text{ and } b \in D)$$

$$\Rightarrow (a \in A \text{ and } a \in C) \text{ and } (b \in B \text{ and } b \in D)$$

$$\Rightarrow a \in (A \cap C) \text{ and } b \in (B \cap D)$$

$$\Rightarrow (a, b) \in (A \cap C) \times (B \cap D).$$

Thus,

$$(A \times B) \cap (C \times D) \subseteq (A \cap C) \times (B \cap D)$$

Similarly,

$$(A \cap C) \times (B \cap D) \subseteq (A \times B) \cap (C \times D)$$

Hence,

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$$



EXAMPLE 1.4 For any non-empty sets A and B prove that $A \times B = B \times A \Leftrightarrow A = B$.

Solution Assume $A = B$. Then, $A \times B = A \times A$ and $B \times A = A \times A$. Thus,

$$A \times B = B \times A$$

Conversely, let $A \times B = B \times A$ and let $x \in A$. Then,

$$x \in A \Rightarrow (x, b) \in A \times B \text{ for } b \in B$$

$$\Rightarrow (x, b) \in B \times A \text{ [since } A \times B = B \times A]$$

$$\Rightarrow x \in B$$

Thus,

$$A \subseteq B$$

Similarly,

$$B \subseteq A$$

Hence,

$$A = B$$



EXAMPLE 1.5 If A and B are two non-empty sets having n elements in common, then prove that $A \times B$ has n^2 elements in common.

Solution Assume $C = A \cap B$. Then,

$$\begin{aligned}
 C \times C &= (A \times B) \cap (B \times A), \text{ since } (a, b) \in C \times C \\
 &\Leftrightarrow a \in C \text{ and } b \in C \Leftrightarrow a \in A \cap B \text{ and } b \in A \cap B \text{ [since } C = A \cap B] \\
 &\Leftrightarrow (a \in A \text{ and } a \in B) \text{ and } (b \in A \text{ and } b \in B) \\
 &\Leftrightarrow (a \in A \text{ and } b \in B) \text{ and } (a \in B \text{ and } b \in A) \\
 &\Leftrightarrow (a, b) \in A \times B \text{ and } (a, b) \in B \times A \\
 &\Leftrightarrow (a, b) \in (A \times B) \cap (B \times A)
 \end{aligned}$$

Thus,

$$C \times C = (A \times B) \cap (B \times A)$$

Again, since $C \times C$ has n^2 elements, so $(A \times B) \cap (B \times A)$ has n^2 elements, i.e., $(A \times B)$ and $(B \times A)$ have n^2 elements in common.



EXAMPLE 1.6 Assume A and B as two sets having two elements in common. If $n(A) = 5$ and $n(B) = 3$, find $n(A \times B)$ and $n\{(A \times B) \cap (B \times A)\}$.

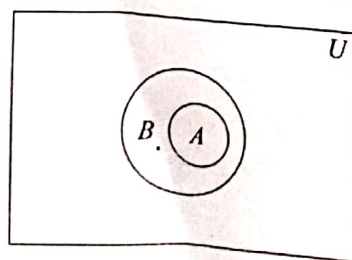
Solution Here, $n(A \times B) = n(A) \cdot n(B) = 5 \cdot 3 = 15$. Since A and B have 2 elements in common, hence $A \times B$ and $B \times A$ have 2^2 elements, i.e., 4 elements in common. Hence, $n\{(A \times B) \cap (B \times A)\} = 4$.

1.4 VENN DIAGRAMS

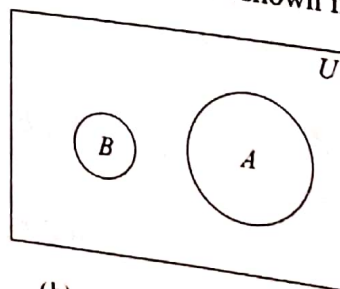
To express the relationship among sets in a perspective way, we represent them (sets) pictorially by means of diagrams, known as *Venn diagrams*.

The universal set is usually represented by a rectangular region and its subsets by closed bounded regions (circles) inside this rectangular region.

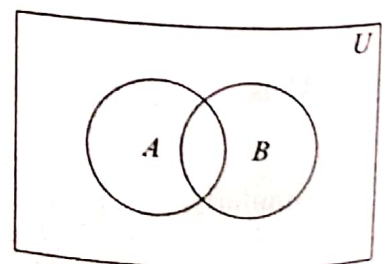
If $A \subseteq B$, then the circle representing A will be entirely within the circle representing B as shown in Figure 1.1(a). If A and B are disjoint, that is if they have no element in common, then the circle representing A will be separated from the circle representing B as in Figure 1.1(b). However, if A and B are two arbitrary sets, it may be possible that some elements are present in A but absent in B , some are present in B but absent in A , some may be neither present in A nor in B . Hence, in general we can represent A and B as shown in Figure 1.1(c).



(a) $A \subseteq B$



(b) A and B are disjoint



(c) A and B

FIGURE 1.1

1.5 OPERATIONS ON SETS

This section presents a number of key operations on sets.

1.5.1 Union of Sets

The union of two sets A and B , denoted by $A \cup B$ (read as 'A union B'), is the set of all those elements, each one of which belongs to A or to B or belongs to both A and B . Symbolically,

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

Here, 'or' is used in the sense of and/or. Figure 1.2 shows a Venn diagram in which $A \cup B$ is shaded.

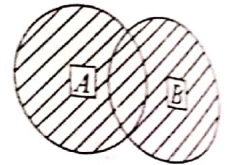


FIGURE 1.2 $A \cup B$

For example

(i) If $A = \{5, 7, 8\}$ and $B = \{2, 7, 9, 10, 11\}$, then
 $A \cup B = \{2, 5, 7, 9, 10, 11\}$.

(ii) If $A = \{x : x \in \mathbb{Z}, x \geq 3\}$ and $B = \{x : x \in \mathbb{Z}, x \geq 8\}$, then
 $A \cup B = \{x : x \in \mathbb{Z}, x \geq 3\}$, where $\mathbb{Z}$ denotes the set of integers.

In general, if $A_1, A_2, A_3, \dots, A_n$ represent sets, then the union of these sets, denoted by

$\bigcup_{i=1}^n A_i$, is expressed as

$$\bigcup_{i=1}^n A_i = \{x : x \in A_i \text{ for at least one set } A_i\}$$

Also, $x \in A \cup B \Rightarrow x \in A \text{ or } x \in B$; and $x \notin A \cup B \Rightarrow x \notin A \text{ and } x \notin B$.

1.5.2 Intersection of Sets

The intersection of two sets A and B , denoted by $A \cap B$, is the set of all elements, common to both A and B . Thus, $A \cap B = \{x : x \in A \text{ and } x \in B\}$.

Figure 1.3 shows a Venn diagram in which $A \cap B$ is shaded.

Also,

$$x \in A \cap B \Rightarrow x \in A \text{ and } x \in B \quad \text{and} \quad x \notin A \cap B \Rightarrow x \notin A \text{ or } x \notin B$$

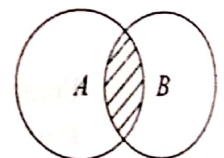


FIGURE 1.3 $A \cap B$

For example

(i) Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6\}$. Then, $A \cap B = \{2, 4\}$.

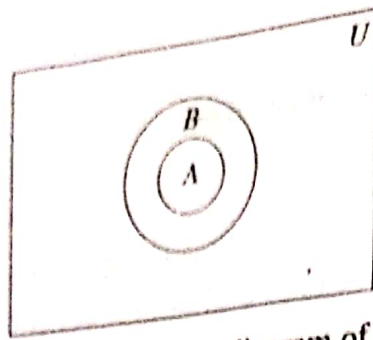
(ii) Let $A = \{x : x = 3n, n \in \mathbb{N}\}$ and $B = \{x : x = 4n, n \in \mathbb{N}\}$.

Then, $A \cap B = \{x : x = 12n, n \in \mathbb{N}\}$.

If $A \cap B = \phi$, i.e., if A and B do not have any common elements, then A and B are said to be *disjoint* or *non-intersecting sets*.

EXAMPLE 1.7 Let A and B be two sets such that $(A \cap B) \subseteq B$ and $B \not\subseteq A$. Draw the corresponding Venn diagram.

Solution Since $(A \cap B) \subseteq B$, it means that every member of A is also a member of B , i.e., $A \subseteq B$. Also, $B \not\subseteq A$ implies that A and B are not equal. Hence, Figure 1.4 shows a Venn diagram in which $A \subseteq B$ is shown.

FIGURE 1.4 Venn diagram of $A \subseteq B$ 

EXAMPLE 1.8 Let A , B , and C be sets such that

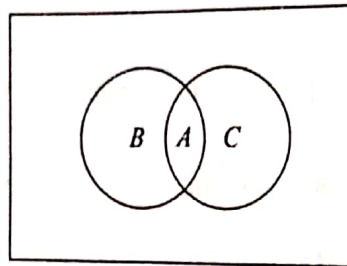
- (i) $A \subseteq B$, $A \subseteq C$, $(B \cap C) \subseteq A$, and $A \subseteq (B \cap C)$
 (ii) $(A \cap B \cap C) = \phi$, $A \cap B = \phi$, $B \cap C = \phi$, and $A \cap C = \phi$

Draw the corresponding Venn diagrams.

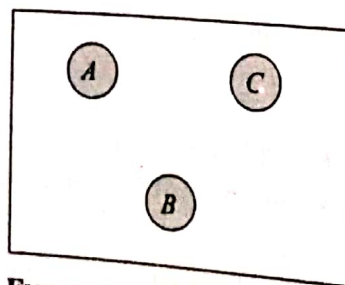
Solution

- (i) Here, $(B \cap C) \subseteq A$ and $A \subseteq (B \cap C)$. This implies that $A = B \cap C$.

Hence, Figure 1.5 shows a Venn diagram in which $A \subseteq B$ and $A \subseteq C$ are shown.

FIGURE 1.5 Venn diagram of $A \subseteq B$ and $A \subseteq C$

- (ii) From the given conditions it can be said that A , B , and C are disjoint sets. This can be shown pictorially using Venn diagram (Figure 1.6).

FIGURE 1.6 Sets A , B , and C are disjoint

1.5.3 Complements

Let U be the universal set and let $A \subset U$. Then the *complement* of A , denoted by A' , is the set of elements which belong to U but do not belong to A , i.e.,

$$A' = \{x : x \in U, x \notin A\}$$

Clearly, $x \in A' \Leftrightarrow x \notin A$. Figure 1.7(a) is a Venn diagram in which A' is shaded.

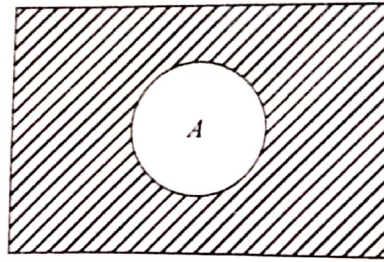
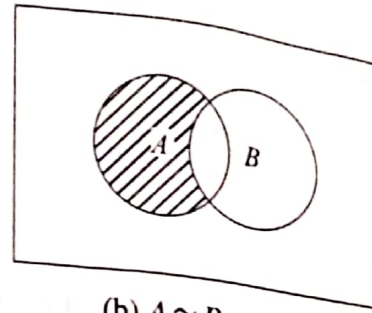
(a) A' (b) $A \sim B$

FIGURE 1.7

For example

- (i) If $U = \{1, 2, 3, 4, 5\}$ and $A = \{2, 4\}$, then $A' = \{1, 3, 5\}$.
- (ii) Let $U = \{x : x \text{ is a letter in English alphabet}\}$ and $A = \{x : x \text{ is a vowel}\}$, then $A' = \{x : x \text{ is a consonant}\}$.

The *relative complement* of a set B with respect to a set A , or simply, the *difference* of A and B , denoted by $A \sim B$, is the set of elements which belong to A , but which do not belong to B , i.e.,

$$A \sim B = \{x : x \in A, x \notin B\}$$

The set $A \sim B$ is read as 'A minus B'. Figure 1.7(b) shows a Venn diagram in which $A \sim B$ is shaded.

1.5.4 Symmetric Difference

The *symmetric difference* of sets A and B , denoted by $A \oplus B$, or $A \Delta B$, is the set of elements which belong to A or B but not to both, i.e.,

$$A \oplus B = (A \cup B) \sim (A \cap B)$$

This may also be expressed as

$$A \oplus B = (A \sim B) \cup (B \sim A)$$

For example If $A = \{-2, 0, 1, 2\}$ and $B = \{1, 2, 3, 4\}$, then

$$A \sim B = \{-2, 0\} \text{ and } B \sim A = \{3, 4\}$$

Thus,

$$A \oplus B = (A \sim B) \cup (B \sim A) = \{-2, 0\} \cup \{3, 4\} = \{-2, 0, 3, 4\}$$

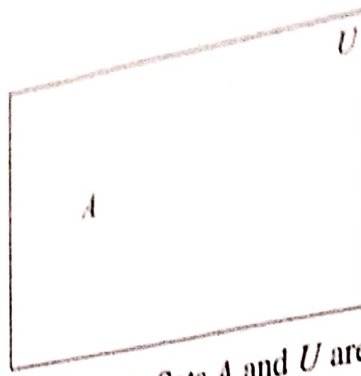


EXAMPLE 1.9 Express the relation which must be valid between sets under the following conditions. Also, draw Venn diagrams.

- (i) $A' \cap U = \phi$
- (ii) $(A \cap B)' = B'$
- (iii) $A \cap B = A \cap C$ and $A' \cap B = A' \cap C$

Solution

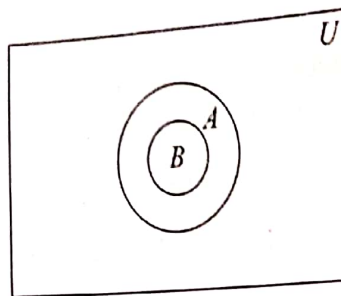
- (i) The given condition shows that A' and U are disjoint sets. Thus, Figure 1.8 shows a Venn diagram in which A and U are equal.

FIGURE 1.8 Sets A and U are equal

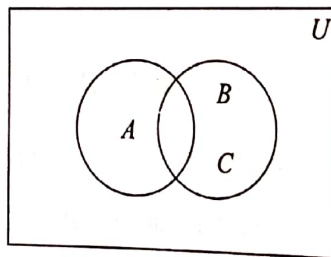
(ii) The left-hand side of the given condition can be rewritten as

$$(A \cap B)' = A' \cup B', \text{ i.e., } A' \cup B' = B'$$

$\Rightarrow B$ must be a subset of A and the Venn diagram for this is shown in Figure 1.9.

FIGURE 1.9 $B \subseteq A$

(iii) The given conditions imply that the sets B and C are equal and its Venn diagram is depicted in Figure 1.10.

FIGURE 1.10 Sets B and C are equal

1.6 COUNTABLE AND UNCOUNTABLE SETS

If a rule is such that it associates with each element $a \in A$, one and only one element $b \in B$, then this rule is called *one-to-one*. Also, if for each element $b \in B$, there exists exactly one element $a \in A$, then this rule is called *one-to-one and onto*, which is also known as *one-to-one correspondence*.

A set which is not finite is called an *infinite set*. A set is said to be *countably infinite*, if there exists one-to-one correspondence between the elements in the set and elements in N ; that is, if the set is countably infinite, we can make a list of its members in such a way that each one corresponds

uniquely to a natural number. A countably infinite set is also termed as *denumerable*. The set of non-negative even integers is countably infinite.

A set which is either finite or denumerable is called *countable*. A set which is not countable is called as *uncountable*. A set which is not countably infinite is called *uncountably infinite* set or *non-denumerable* set or simply *uncountable* set.

For example The most accomplished example of an uncountable set is the set of all real numbers that can be represented by an infinite decimal of the form $0.a_1a_2a_3\ldots$, where a_i is an integer satisfying the inequality $0 \leq a_i \leq 9$.



EXAMPLE 1.10 Investigate on finite, countably infinite, or uncountable sets with respect to one-to-one correspondence between the set of N and the following sets:

- The set of even integers
- The set of negative integers
- The set of integers that are multiples of 5
- The set of real numbers between 0 and 0.6
- The set of all trees on earth

Solution

- (i) The set of even integers is countably infinite as we can form one-to-one correspondence with N as,

1	2	3	4	5	6	7	...
↕	↕	↕	↕	↕	↕	↕	
0	-2	2	-4	4	-6	6	...

- (ii) The set of negative integers is countably infinite, as we can form one to one correspondence with N as,

1	2	3	4	5	6	7	...
↕	↕	↕	↕	↕	↕	↕	
-1	-2	-3	-4	-5	-6	-7	...

- (iii) The set of integers that are multiples of 5 is countably infinite, as the elements of above-mentioned set has one-to-one correspondence with elements in N ,

1	2	3	4	5	6	7	8	...
↕	↕	↕	↕	↕	↕	↕	↕	
0	5	10	15	20	25	30	35	...

- (iv) The set of real numbers between 0 and 0.6 is uncountable.
 (v) The set of all trees on earth, even though the number is large, is finite.



EXAMPLE 1.11 State whether the following sets are finite, countably finite, or uncountably finite.

- Set of all prime numbers
- Set of real numbers between 0 and 1
- Number of fish in Indian ocean

Solution

- (i) Countably infinite
- (ii) Uncountably infinite
- (iii) Finite, even though the count is too large



EXAMPLE 1.12 Illustrating with an example, show that the intersection of countably infinite number of infinite sets may be finite.

Solution

Suppose $A =$ Set of all integers less than or equal to 2

$$A = \{2, 1, 0, -1, -2, -3, \dots\}$$

$$N = \{\text{Set of natural numbers}\}$$

$$= \{1, 2, 3, \dots\}$$

Thus, $A \cap N = \{1, 2\}$ is finite. Hence, intersection of two countably infinite sets could be finite.

1.7 ALGEBRA OF SETS

There exists a number of relationships among set operations. These operations satisfying the laws or identities form the basis of the algebraic laws of set operations. Sets under the operations of union, intersection, and complement also satisfy various laws or identities which are listed in Table 1.1.

There are two ways of tackling the proofs of the identities involving set operations. One way to show that the two sets are equal is to illustrate that each is a subset of the other.

Recall that to show that one set is a subset of a second set, we can prove that if an element belongs to the first set, then it must also belong to the second set. We will use here a direct proof. On the other way we can use Venn diagrams. The proofs of some of the laws or identities given in Table 1.1 will be presented here and the remaining is left to the reader.

TABLE 1.1 Set identities

Name	Identity
I. Identity laws	$A \cup \phi = A$ $A \cap U = A$
II. Domination laws	$A \cup U = U$ $A \cap \phi = \phi$
III. Idempotent laws	$A \cup A = A$ $A \cap A = A$
IV. Complement laws	$A \cup A' = U$ $A \cap A' = \phi$
V. Involution law or complementation laws	$(A')' = A$
VI. Commutative laws	$A \cup B = B \cup A$ $A \cap B = B \cap A$
VII. Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
VIII. Distributive laws	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
IX. De Morgan's laws	$(A \cup B)' = A' \cap B'$ $(A \cap B)' = A' \cup B'$
X. Absorption laws	$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$

1.8 MULTISSET

A set is inherently an unordered collection of distinct elements. This unordered collection matters in case of repetitive occurrence of elements. For instance, the repetition occurs when we count each mark obtained by the number of students in an examination, i.e., in case of counting of frequency of numbers. This typical situation can be tackled by the concept like multiset.

Multisets are unordered collection of elements in which an element can occur as a member more than once (or repeatedly). Hence, multiset is a set in which elements are not necessarily distinct.

For example Let $A = \{1, 1, 1, 2, 2, 3\}$. Here, 1 appears three times, 2 appears twice, and 3 appears once in the set. Also, $\{3, 3, 3\}$, $\{x, y, z\}$, and $\{a, a, a, b, b, c, c, d\}$ are the multisets.

The notation used to represent multiset is given by

$$S = \{m_1 \cdot a_1, m_2 \cdot a_2, \dots, m_n \cdot a_n\}$$

in which the element a_1 appears m_1 times, element a_2 appears m_2 times, and so on. The numbers $m_i, i = 1, 2, \dots, n$ are called the *multiplicities* of the elements a_i . Actually, the *multiplicity* of an element in a multiset is defined to be the number of items the element appears in the multiset.

Note A set is a special case of multiset in which multiplicity of each element is either 0 or 1.

1.8.1 Operations on Multisets

Union of Multisets

Let A and B be two multisets. Then, the *union* of the multisets A and B is the multiset where the multiplicity of an element is the maximum of its multiplicities in A and B .

Let $A = \{a, a, b, b, c\}$ and $B = \{a, a, b, d\}$. Then,

$$A \cup B = \{a, a, b, b, c, d\}$$

Intersection of Multisets

The *intersection* of A and B is the multiset in which the multiplicity of an element is the minimum of its multiplicities in A and B , i.e.,

$$A \cap B = \{a, a, b\}$$

Sum of Multisets

The *sum* of A and B , denoted by $A + B$, is the multiset in which the multiplicity of an element is the sum of multiplicities in set A and set B .

Let $A = \{a, a, b, c, c\}$ and $B = \{a, b, b, d\}$. Then,

$$A + B = \{a, a, a, b, b, b, c, c, d\}$$

Difference of Multisets

The *difference* of A and B , denoted by $A - B$, is the multiset in which the multiplicity of an element equals to the multiplicity of element in A minus the multiplicity of element in B if the difference is positive, and is equal to zero if the difference is zero and negative.

Let $A = \{a, a, b, c, c\}$ and $B = \{a, b, b, d\}$. Then,

$$A - B = \{a, c, c\}$$



EXAMPLE 1.13 Let A and B be two multisets defined as $A = \{2 \cdot a, 3 \cdot b, 1 \cdot c\}$ and $B = \{1 \cdot a, 4 \cdot b, 1 \cdot d\}$, respectively. Then, determine

- (i) $A \cup B$ (ii) $A \cap B$ (iii) $A + B$ (iv) $A - B$ (v) $B - A$

Solution

- (i) $A \cup B = \{2 \cdot a, 4 \cdot b, 1 \cdot c, 1 \cdot d\}$
 (ii) $A \cap B = \{2 \cdot a, 3 \cdot b\}$
 (iii) $A + B = \{3 \cdot a, 7 \cdot b, 1 \cdot c, 1 \cdot d\}$

- (iv) $A - B = \{1 \cdot a, 1 \cdot c\}$
 (v) $B - A = \{1 \cdot b, 1 \cdot d\}$

Inclusion Exclusion Principle

Theorem 4. If A , B and C be finite sets then

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$$

for finite sets A and B , $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Or

If A and B are finite sets, then [Raj. 09; Ajmer 09; Alwar 11; Silke

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example 44. In a class containing 50 students, 15 play Tennis, 20 play Cricket and 20 play Hockey, 3 play Tennis and Cricket, 6 play Cricket and Hockey, and 5 play Tennis and Hockey. 7 play no game at all. How many play Cricket, Tennis and Hockey.

Solution : Let the sets of Tennis, Cricket and Hockey players be denoted by T , C and H respectively then $n(U) = 50$, $n(T) = 15$, $n(C) = 20$, $n(H) = 20$, $n(T \cap C) = 3$, $n(C \cap H) = 6$, $n(T \cap H) = 5$, $n(T \cup C \cup H)' = 7$.

We want to find $n(T \cap C \cap H)$.

$$\text{Now } n(T \cup C \cup H) = n(T) + n(C) + n(H) - n(T \cap C) - n(C \cap H) - n(T \cap H) + n(T \cap C \cap H)$$

$$\therefore n(T \cup C \cup H) = 15 + 20 + 20 - 3 - 6 - 5 + n(T \cap C \cap H)$$

$$\text{or } n(T \cup C \cup H) = 41 + n(T \cap C \cap H)$$

$$\text{But } n(T \cup C \cup H)' = n(U) - n(T \cup C \cup H)$$

$$\therefore 7 = 50 - [41 + n(T \cap C \cap H)]$$

$$\text{or } n(T \cap C \cap H) = 2.$$

Example 45. In a class of 50 students, 30 are studying Hindi and 25 English language and 10 are studying both languages. How many students are studying either language ?

Solution : Let U be the class of 50 students, A is a set of those students studying Hindi and B be the set of those studying English. We have to find $n(A \cup B)$

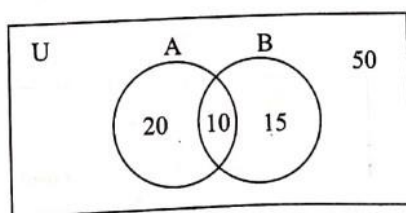


Fig. 16

Given that $n(A) = 30$, $n(B) = 15$, $n(A \cap B) = 10$

we have,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$\therefore n(A \cup B) = 30 + 15 - 10 = 35$$

Therefore, the number of students who studying either of languages are 35.

Example 49. Let 100 of the 120 students of Mathematics at a college take at least one of the languages Hindi, English and German. Also, let 65 study Hindi, 45 study English and 42 German. If 20 study Hindi and English, 25 study English and German and 15 study Hindi and German. Find the number of students who study all the three languages.

[RTU 2011, 200]

Solution. Let A, B, C denote the set of students who study Hindi, English and German language, respectively. Then given that,

$$|A \cup B \cup C| = 100, |A| = 65, |B| = 45, |C| = 42,$$

$$|A \cap B| = 20, |B \cap C| = 25, |A \cap C| = 15.$$

We have to find $|A \cap B \cap C|$.

By Principle of Inclusion – Exclusion, we have

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$\Rightarrow 100 = 65 + 45 + 42 - 20 - 15 - 25 + |A \cap B \cap C|$$

$$\Rightarrow |A \cap B \cap C| = 160 - 152 = 8$$

Hence 8 students study all the three languages.

Example 50. In a group of 52 persons 16 drink tea but not coffee and 33 drink tea. Find

(i) how many drink tea and coffee both.

(ii) how many drink coffee but not tea.

[Raj. 2005]

Solution : Let U be the set of 52 persons. Let T denotes the set of persons who drink tea and C denotes the set of persons who drink coffee.

Let number of person who drink Tea but not Coffee i.e., $n(T - C) = x_1$

number of person who drink Tea and Coffee both i.e., $n(T \cap C) = x_2$

number of person who drink Coffee but not Tea, i.e., $n(C - T) = x_3$

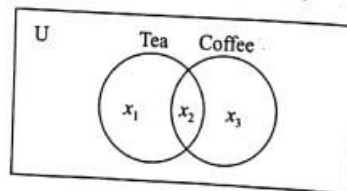


Fig. 20

then, we have

$$x_1 + x_2 + x_3 = 52$$

$$x_1 = 16$$

$$x_1 + x_2 = 33$$

$$\therefore x_2 = 33 - 16 = 17 \text{ and } x_3 = 52 - 33 = 19$$

$$\therefore \text{(i) Those who drink tea and coffee both} = x_2 = 17$$

$$\text{(ii) Those who drink coffee but not tea} = x_3 = 19$$

divisible by 9.
§1.12 आविष्टि-अपवर्जन सिद्धान्त (Principle of Inclusion-Exclusion) [Raj. 09; Ajmer 09; Alwar 11]

यदि समुच्चय असंयुक्त हैं, तब इन असंयुक्त समुच्चयों के संघ में उपस्थित अवयवों की गणना, योग नियम (sum rule) द्वारा आसानी से की जा सकती है। परन्तु यदि समुच्चय असंयुक्त नहीं हैं तो योग नियम में कुछ संशोधन करने पर जिस नियम की उत्पत्ति होती है उसे आविष्टि-अपवर्जन सिद्धान्त (Principle of inclusion-exclusion) कहते हैं।

प्रमेय 1.12.1. यदि A तथा B दो परिमित समुच्चय हैं, तब

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

If A and B are finite sets, then [Raj. 09; Ajmer 09; Alwar 11; Sikar]

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

उपपत्ति : यदि A और B कोई दो समुच्चय हैं, तो हम जानते हैं कि,

$$A \cup B = (A \cap B') \cup (A \cap B) \cup (A' \cap B) \text{ जहाँ } A \cap B', A \cap B$$

और $A' \cap B$ परस्पर असंयुक्त समुच्चय हैं।

अतः योग नियम से,

$$|A \cup B| = |A \cap B'| + |A \cap B| + |A' \cap B| \quad \dots (1)$$

पुनः $A = (A \cap B') \cup (A \cap B)$

$$\therefore |A| = |A \cap B'| + |A \cap B| \quad \dots (2)$$

इसी प्रकार चूँकि $B = (A \cap B) \cup (A' \cap B)$

$$\therefore |B| = |A \cap B| + |A' \cap B| \quad \dots (3)$$

(2) तथा (3) से,

$$|A| + |B| = |A \cap B'| + |A \cap B| + |A \cap B| + |A' \cap B|$$

$$\text{या } |A| + |B| - |A \cap B| = |A \cap B'| + |A \cap B| + |A' \cap B| \quad \dots (4)$$

(1) तथा (4) से स्पष्ट है कि

$$|A \cup B| = |A| + |B| - |A \cap B|$$

उदाहरण. 12 विषयों के एक समूह में से किन्हीं 5 विषयों का संचय कितने प्रकार से बनाया जा सकता है कि प्रत्येक संचय में गणित और सांख्यिकी में से कम से कम एक विषय अवश्य रहे ?

From a group of 12 subjects, how many combinations of 5 subjects can be formed so that at least one subject in mathematics and statistics always remains there ?

हल : माना 12 विषयों में से किन्हीं 5 विषयों के कुल ${}^{12}C_5 = 792$ संचयों A तथा B उन संचयों का समुच्चय हैं जिनमें क्रमशः गणित विषय तथा सांख्यिकी विषय सम्मिलित हैं।

तब, $|A| = {}^{11}C_4 = |B|$ अर्थात् $|A| = 330 = |B|$

तथा $|A \cap B| = {}^{10}C_3 = 120$

अतः आविष्टि-अपवर्जन सिद्धान्त से,

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$= 330 + 330 - 120 = 540$$

अतएव 540 ऐसे संचय होंगे जिनमें कम से कम गणित तथा सांख्यिकी विषय अवश्य होगा।

प्रमेय 1.12.2. यदि A, B और C तीन परिमित समुच्चय हैं। तब (If A, B and C are three finite sets, then) [Sikar 18] Alwar 17Raj. 10]

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

उपपत्ति : माना A, B और C तीन परिमित समुच्चय हैं। तब इन्हें असंयुक्त उप समुच्चयों के रूप में निम्न प्रकार से व्यक्त किया जा सकता है :

$$A = (A \cap B' \cap C') \cup (A \cap B \cap C') \cup (A \cap B' \cap C) \cup (A \cap B \cap C),$$

$$B = (A' \cap B \cap C') \cup (A \cap B \cap C') \cup (A' \cap B \cap C) \cup (A \cap B \cap C),$$

तथा, $C = (A' \cap B' \cap C) \cup (A \cap B' \cap C) \cup (A' \cap B \cap C) \cup (A \cap B \cap C)$

28]

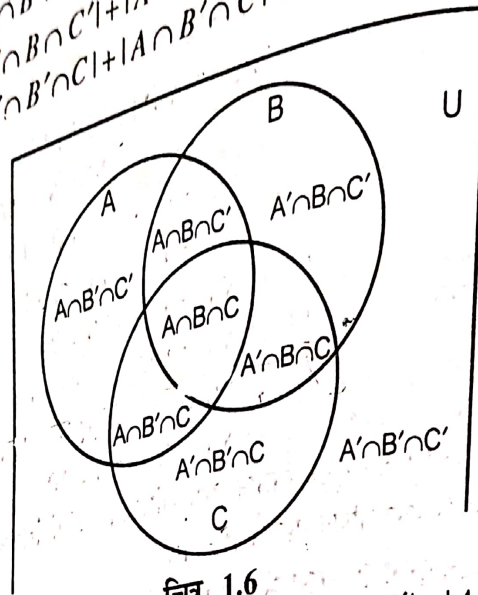
चूँकि 'उपरोक्त सभी समुच्चय $A \cap B' \cap C', A \cap B \cap C', A \cap B' \cap C$ इत्यादि परस्पर असंयुक्त हैं, अतः योग नियम से,

$$|A| = |A \cap B' \cap C'| + |A \cap B \cap C'| + |A \cap B' \cap C| + |A \cap B \cap C|$$

$$|B| = |A' \cap B \cap C'| + |A \cap B \cap C'| + |A' \cap B \cap C| + |A \cap B \cap C|$$

$$|C| = |A' \cap B' \cap C| + |A \cap B' \cap C| + |A' \cap B \cap C| + |A \cap B \cap C|$$

तथा



चित्र 1.6

$$\begin{aligned}
 |A| + |B| + |C| &= [|A \cap B' \cap C'| + |A \cap B \cap C'| + |A \cap B' \cap C| + |A \cap B \cap C|] \\
 &+ [|A' \cap B \cap C'| + |A \cap B \cap C'| + |A' \cap B \cap C| + |A \cap B \cap C|] \\
 &+ [|A' \cap B' \cap C| + |A \cap B' \cap C| + |A' \cap B \cap C| + |A \cap B \cap C|] - |A \cap B \cap C| \\
 &= |A \cup B \cup C| + |A \cap B| + |A \cap C| + |B \cap C| - |A \cap B \cap C| \\
 \text{या, } |A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|
 \end{aligned}$$

उदाहरण 1. 90 विद्यार्थियों के एक समूह में, जिसमें प्रत्येक विद्यार्थी के पास कम से कम गणित, कम्प्यूटर विज्ञान या सांख्यिकी का विषय है, यह पाया गया कि 40 विद्यार्थियों के पास गणित, 50 विद्यार्थियों के पास कम्प्यूटर विज्ञान तथा 60 विद्यार्थियों के पास सांख्यिकी विषय हैं। 10 विद्यार्थियों के पास गणित तथा कम्प्यूटर विज्ञान, 40 के पास गणित तथा सांख्यिकी तथा 10 विद्यार्थियों के पास सभी तीन विषय हैं। उन विद्यार्थियों की संख्या ज्ञात कीजिए जिनके पास कम्प्यूटर विज्ञान तथा सांख्यिकी विषय हैं परन्तु गणित नहीं है।

In a group of 90 students, each of whom has taken at least Mathematics or Computer science or Statistics, it were found 40 students having Mathematics, 50 students having Computer science and 60 students having statistics. 10 students have Mathematics as well as computer science, 40 students have Mathematics as well as statistics and 10 students have all the three subjects. Find the number of students who have computer science and statistics but not Mathematics.

हल : माना M , C और S , क्रमशः गणित, कम्प्यूटर विज्ञान और सांख्यिकी विषय लेने वाले विद्यार्थियों के समुच्चय हैं। तब दिया है कि,

$$|M \cup C \cup S| = 90; |M| = 40; |C| = 50; |S| = 60$$

$$|M \cap C| = 10; |M \cap S| = 40 \text{ तथा } |M \cap C \cap S| = 10$$

हमें ज्ञात करना है $|M' \cap C \cap S|$

आविष्टि और अपवर्जन के सिद्धान्त से,

$$|M \cup C \cup S| = |M| + |C| + |S| - |M \cap C|$$

$$- |M \cap S| - |C \cap S| + |M \cap C \cap S|$$

$$\therefore 90 = 40 + 50 + 60 - 10 - 40 - |C \cap S| + 10$$

$$= 110 - |C \cap S|$$

$$\therefore |C \cap S| = 110 - 90 = 20$$

परन्तु $|C \cap S| = |M \cap C \cap S| + |M' \cap C \cap S|$

$$\therefore 20 = 10 + |M' \cap C \cap S|$$

या, $|M' \cap C \cap S| = 20 - 10 = 10.$

अतएव कम्प्यूटर विज्ञान तथा सांख्यिकी दोनों विषय लेने वाले विद्यार्थियों की संख्या (जिनके पास गणित नहीं है) = 10.

उदाहरण 2. चेन्नई में किसी कॉलेज के विद्यार्थियों का भाषा सर्वेक्षण करने पर पाया गया कि 900 विद्यार्थी अंग्रेजी, 1200 विद्यार्थी तमिल तथा 750 विद्यार्थी हिन्दी जानते हैं। 450 विद्यार्थी अंग्रेजी तथा तमिल, 300 विद्यार्थी अंग्रेजी तथा हिन्दी और 200 विद्यार्थी तमिल तथा हिन्दी का ज्ञान रखते हैं। 150 विद्यार्थी तीनों भाषाओं को जानते हैं। बताइए कि कितने विद्यार्थी [Raj. 18] [Subodh Auto. 2014]

(i) केवल हिन्दी

(ii) कम से कम एक भाषा

(iii) अंग्रेजी तथा एक अन्य भाषा तमिल और हिन्दी में से केवल एक परन्तु दोनों नहीं)

(iv) कम से कम दो भाषाएँ जानते हैं।

In a language survey of students of a College in Chennai, it was found that 900 students know English, 1200 students know Tamil and 750 students know Hindi. 450 students know English and Tamil, 300 students know English and Hindi while 200 students know Tamil and Hindi. 150 students know all the three languages. Find how many students know :

(i) Hindi only

(ii) at least one language

(iii) English and one but not both of Tamil and Hindi

(iv) at least two languages.

हल : (i) माना E , T तथा H क्रमशः अंग्रेजी, तमिल तथा हिन्दी जानने वाले विद्यार्थियों के समुच्चय हैं। वैन आरेख का उपयोग करने पर समस्या को अत्यन्त आसानी से हल किया जा सकता है।

वैन आरेख से स्पष्ट है कि :

(i) केवल हिन्दी का ज्ञान रखने वाले विद्यार्थियों की संख्या = 400

(ii) कम से कम एक भाषा का ज्ञान रखने वाले विद्यार्थियों की संख्या

$$= 300 + 700 + 400 + 150$$

$$+ 300 + 50 + 150 = 2050$$

(iii) अंग्रेजी तथा तमिल और हिन्दी में से कोई एक भाषा का ज्ञान रखने वाले विद्यार्थियों की संख्या = $300 + 150 = 450$

(iv) कम से कम दो भाषाओं का ज्ञान रखने वाले विद्यार्थियों की संख्या

$$= 150 + 300 + 50 + 150$$

$$= 650$$

परन्तु यदि हम आविष्टि और अपवर्जन के सिद्धान्त से समस्या का हल करें तो दिया है कि,

$$|E| = 900; |T| = 1200; |H| = 750$$

$$|E \cap T| = 450; |E \cap H| = 300; |H \cap T| = 200$$

$$\text{तथा } |E \cap T \cap H| = 150$$

$$\begin{aligned} \text{अतः } |E \cup T \cup H| &= |E| + |T| + |H| - |E \cap T| - |E \cap H| \\ &\quad - |T \cap H| + |E \cap T \cap H| \\ &= 900 + 1200 + 750 - 450 - 300 - 200 + 150 \\ &= 2050 \end{aligned}$$

अतः कम से कम एक भाषा को जानने वाले विद्यार्थियों की संख्या = 2050

अब यदि

$$P = E \cap H, Q = T \cap H \text{ तथा } R = E \cap T$$

तब

$$|P \cup Q| = |P| + |Q| - |P \cap Q|$$

$$= 300 + 200 - 150$$

$$= 350$$

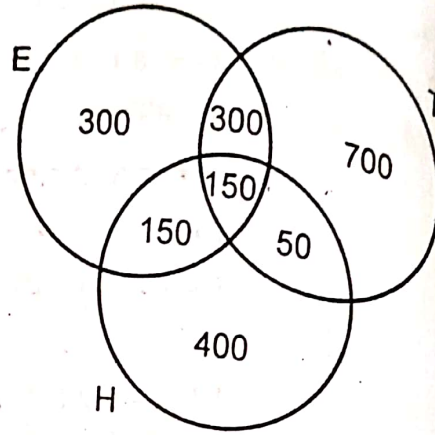
$$[\because P \cap Q = (E \cap H) \cap (T \cap H)]$$

अतः केवल हिन्दी का ज्ञान रखने वाले विद्यार्थियों की संख्या

$$= |H - (P \cup Q)| = |H| - |P \cup Q|$$

$$= 750 - 350 = 400$$

अब अंग्रेजी तथा हिन्दी और तमिल में से कोई एक भाषा का ज्ञान रखने वाले विद्यार्थी



चित्र 1.7

 CamScanner

उदाहरण 1. 1 तथा 1000 के मध्य ऐसे कितने पूर्णांक हैं जो 2, 3, 5 अथवा 7 से विभाजित नहीं होते ?

How many integers are there between 1 and 1000 which are not divisible by 2, 3, 5 or 7 ? [Subodh 14; Ajmer 10; Sikar 16]

हल : माना S , उन सभी पूर्णाकों x का समुच्चय है जहाँ $1 \leq x \leq 1000$. माना A_1, A_2, A_3 तथा A_4 समुच्चय S के उन अवयवों के समुच्चय हैं जो क्रमशः 2, 3, 5 तथा 7 से विभाजित किये जा सकते हैं। तब $A_1 \cap A_2, S$ के उन पूर्णाकों का समुच्चय है जो, 6 से विभाजित होते हैं। हमें $|A_1' \cap A_2' \cap A_3' \cap A_4'|$ ज्ञात करना है।

$$\text{अब } |A_1| = \frac{1000}{2} = 500; |A_2| = \left\lfloor \frac{1000}{3} \right\rfloor = 333$$

[जहाँ $\lfloor x \rfloor$ का अर्थ है महत्तम पूर्णांक $\leq x$]

$$|A_3| = \frac{1000}{5} = 200; |A_4| = \left\lfloor \frac{1000}{7} \right\rfloor = 142;$$

$$|A_1 \cap A_2| = \left\lfloor \frac{1000}{6} \right\rfloor = 166; |A_1 \cap A_3| = \frac{1000}{10} = 100,$$

$$|A_1 \cap A_4| = \left\lfloor \frac{1000}{14} \right\rfloor = 71; |A_2 \cap A_3| = \left\lfloor \frac{1000}{15} \right\rfloor = 66,$$

$$|A_2 \cap A_4| = \left\lfloor \frac{1000}{21} \right\rfloor = 47; |A_3 \cap A_4| = \left\lfloor \frac{1000}{35} \right\rfloor = 28,$$

Function:-

Let X & Y be two non empty sets. ①
A function 'f' is a rule which associates to every element $x \in X$ to unique element of $y \in Y$, then such a rule 'f' is known as a function from the set X to the set Y .

where set X is called the domain of f and Y is called the co-domain of f . The set of all images is called the range of f and is denoted by $f(X)$.

We have -

$$f(X) = \text{Range of } f = \{f(x) : x \in X\}$$

Kinds of functions:-

- ① Equal function:- Two $f: X \rightarrow Y$ & $g: X \rightarrow Y$ are said to be equal if $f(x) = g(x) \forall x \in X$.
- ② One-One function:- A $f: X \rightarrow Y$ is said to be one-one if $f(x) = f(y) \Rightarrow x = y \forall x, y \in X$.
- e.g. $f: \mathbb{N} \rightarrow \mathbb{N}$
 $f(x) = 2x$
if $f(x_1) = f(x_2) \Rightarrow 2x_1 = 2x_2 \Rightarrow x_1 = x_2 \forall x_1, x_2 \in \mathbb{N}$.

- ③ Many-One function:- A function $f: X \rightarrow Y$ is said to be many one function if two or more elements in X have the same image in Y .
- e.g. $f: \mathbb{R} \rightarrow \mathbb{R}$
 $f(x) = x^2$
 $\exists x_1 \neq x_2 \in \mathbb{R}$ s.t.
 $f(x_1) = f(x_2)$
 $f(1) = f(-1)$

(iii) Onto function. ② A function $f: X \rightarrow Y$ is said to be onto function if for each $y \in Y$ $\exists x \in X$ s.t.

$f(x) = y$
i.e. Range of $f = \text{Co-domain of } f$.

$$f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x+2 \quad \forall x \in \mathbb{R}$$

Let y be any element of $\mathbb{R}$ and let x be its pre-image under f . Then we have.

$$f(x) = y \Rightarrow x+2 = y \Rightarrow x = y-2 \in \mathbb{R}$$

$\forall y \in \mathbb{R}, \exists x \in \mathbb{R}$ s.t. $f(x) = y$

$\Rightarrow f$ is onto.

(iv) One-One Onto function (bijection) :-

A one-one onto function is said to be bijection.

(v) Identity function :- $f: X \rightarrow X$ if $f(x) = x$ $\forall x \in X$.

(vi) Constant function :- if $\forall x \in X$ has same image.
i.e. $f(x) = c$ $\forall x \in X$.

(vii) Into function :- if $\exists$ at least one element $y \in Y$ which has no pre-image under f .

$$f: \mathbb{R} \rightarrow \mathbb{R} \\ f(x) = x^2$$

~~$x \in \mathbb{R}$ s.t.~~ $x^2 = -1 \Rightarrow$ there is no $x \in \mathbb{R}$ s.t. $x^2 = -1$, therefore this

function is into.

Composite function:

Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be two functions
s.t. $f(x) = y$ and $g(y) = z$ where $x \in X, y \in Y, z \in Z$.

then a function $h: X \rightarrow Z$ s.t.

$$h(x) = z = g(y) = g(f(x)) \quad \forall x \in X.$$

is called composite function of f & g . and it
is denoted by $g \circ f(x) = g(f(x)) \quad \forall x \in X$.

Some special function:

① Polynomial function:

polynomial functions are functions that
have the form:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0.$$

where n is non-negative integers, and $a_0, \dots, a_n$ are
real no.

② Exponential function:

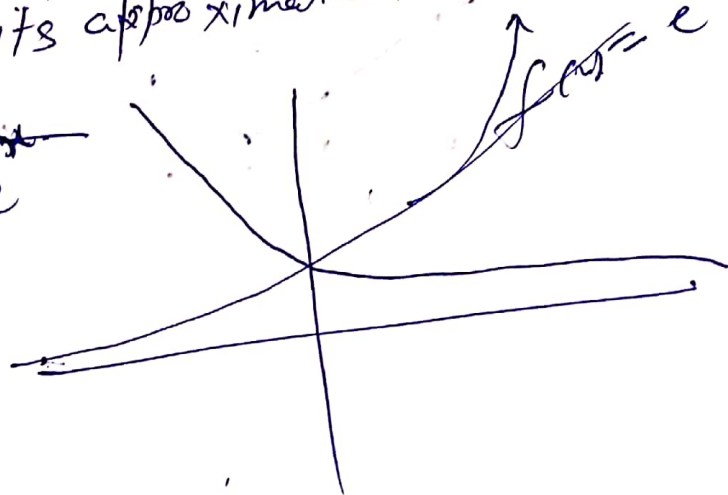
A function of the form

$$f(x) = e^x$$

is said to be exponential function where e
is number ≈ 2.718281828 its approximated value is.

$$f(x) = e$$

$$f(x) = e$$



③ Logarithmic function :- (u)

as, $f(x) = \log_b x$

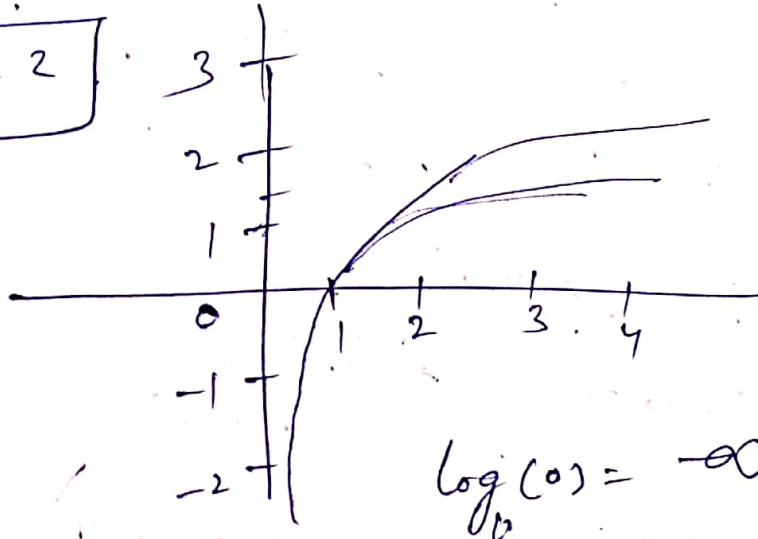
Here, b is the base of the logarithm function.

The logarithmic function is defined as

$$y = f(x) = e^x$$

$\Rightarrow x = \log y$
natural logarithm function is the inverse function of $f(x)$.

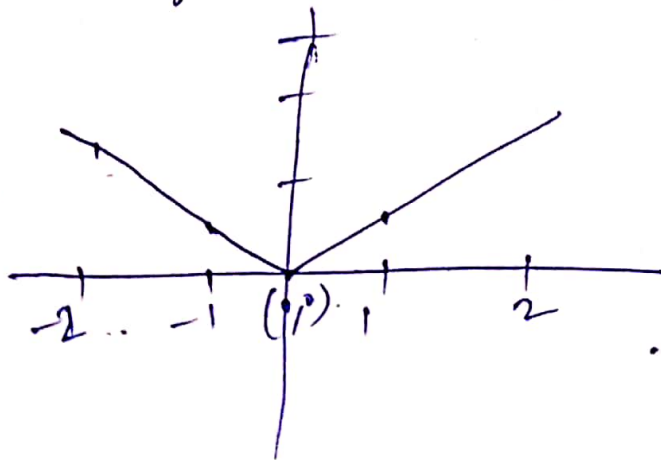
e.g. $3^2 = 9$
 $\log_3 9 = 2$



④ Modulus function :-

for any real no. x , the modulus of x is denoted by $|x|$ and is defined as —

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

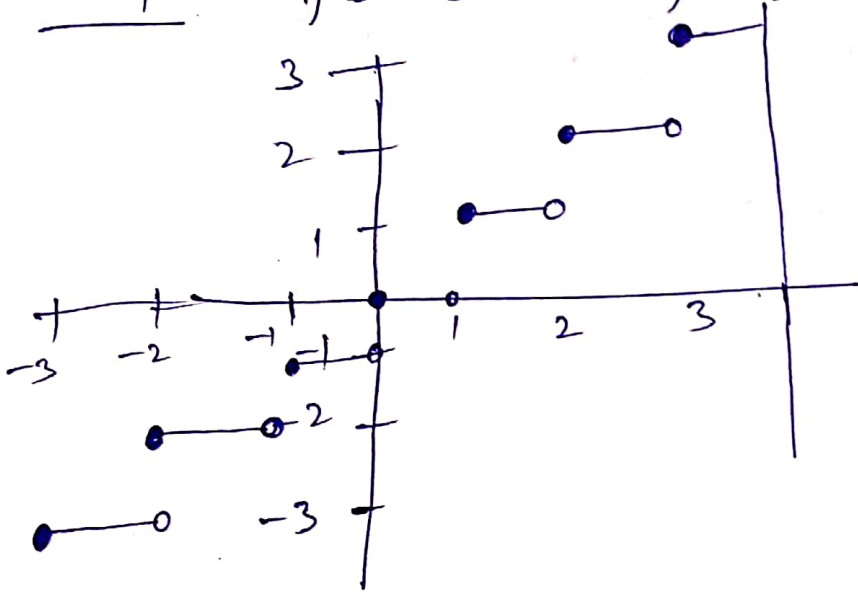


floor function:

The floor function of a real number denoted by $\lfloor x \rfloor$ or $\text{floor}(x)$ is a function that returns the highest integer less than or equal to x .
i.e.

$$\lfloor x \rfloor = \max \{ n \in \mathbb{Z}, n \leq x \}$$

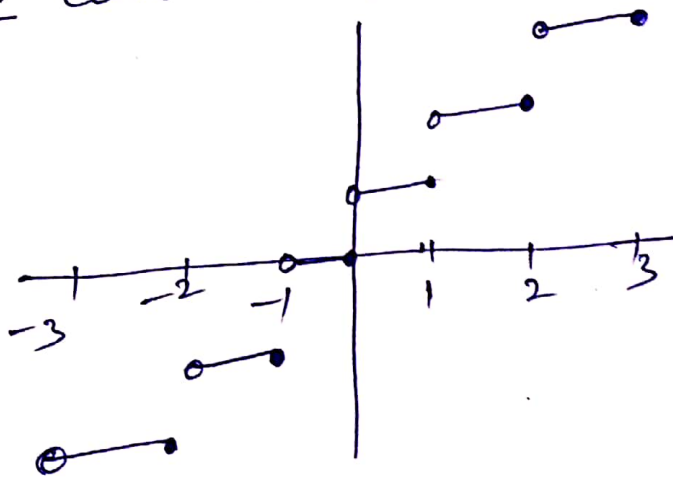
e.g. $\text{floor}(2.9) = 2$, $\text{floor}(-2) = -2$, $\text{floor}(2.3) = -3$.



⑥ Ceiling function: $\text{ceil}(x)$ is the function that returns the smallest integer not less than x . i.e.

$$\lceil x \rceil = \min \{ n \in \mathbb{Z} : x \leq n \}$$

e.g. $\text{ceil}(2.3) = 3$, $\text{ceil}(2) = 2$, $\text{ceil}(-2.3) = -2$.



⑤ mod function:-

⑥

$f(n, y) = n \bmod y$ denotes the remainder when integer x is divided by a +ve integer y .

i.e. $f(n, y) = 24 \bmod 5$
 $= 4$.

⑥ div function:-

$g(n, y) = n \operatorname{div} y$ denotes the quotient when n is divided by y .

e.g. $g(n, y) = 34 \operatorname{div} 5$
 $= 6$.

Equivalent Sets:- Two sets A & B are said to be equivalent if $\exists$ a one-one onto mapping from A to B. It is denoted by $A \sim B$.

e.g. $f: \mathbb{N} \rightarrow \mathbb{E}$
 $f(n) = 2n \quad \forall n \in \mathbb{N}$.

Finite set:- A set is called finite if it contains n -elements. or it is ~~equal~~ equivalent to a set having n -elements.
If a set is not finite, then it is called infinite.

Countable Set:- A set is said to be countable if either A is finite or A is denumerable.
or
if either A is finite or $A \sim \mathbb{N}$.

Denumerable Set:- A set A is said to be denumerable if $\exists$ a one-one mapping from a set $\mathbb{N}$ onto the set A.
i.e. $A \sim \mathbb{N}$ or $\mathbb{N} \sim A$.

Note:- If a set is not countable, then it is said to be uncountable.

e.g. (1) Set of natural no.

$$f: \mathbb{N} \rightarrow \mathbb{N}$$
$$f(n) = n.$$

(2) $f: \mathbb{N} \rightarrow \mathbb{Z}$

$$f(n) = \begin{cases} \frac{n-1}{2}, & n \text{ is odd} \\ -\frac{n}{2}, & n \text{ is even.} \end{cases}$$

to check one-one and onto. {range of f = Co-domain of f }

If A & B are finite $\Rightarrow A \cup B$ finite.

If A & B are finite $\Rightarrow A \cap B$ finite.

Every subset of a countable set is countable.

Every infinite set has a countable subset.

proof! Let A be infinite. Let $a_1 \in A$.

Since A is infinite $\Rightarrow A \neq \{a_1\}$.

Hence $\exists a_2 \neq a_1$ s.t. $a_2 \in A$, since A is infinite.
 $\Rightarrow A \neq \{a_1, a_2\}$.

So $\exists a_3 \neq a_2 \neq a_1$ s.t. $a_3 \in A$

proceeding like this, we obtain a subset

$$B = \{a_1, a_2, a_3, \dots\} \text{ of } A.$$

Define a map $f: \mathbb{N} \rightarrow B$

$$f(k) = a_k \quad \forall k \in \mathbb{N}.$$

Since f is 1-1 and onto $\Rightarrow B$ is countable.

of rational numbers is countable.
 let A_n be a set of rational no. which can be written as —

$$A_1 = \left\{ \frac{0}{1}, -\frac{1}{1}, \frac{1}{1}, -\frac{2}{1}, \frac{2}{1}, \dots \right\}$$

$$A_2 = \left\{ \frac{0}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{2}{2}, \frac{2}{2}, \dots \right\}$$

$$A_n = \left\{ \frac{0}{n}, -\frac{1}{n}, \frac{1}{n}, -\frac{2}{n}, \frac{2}{n}, \dots \right\}$$

then set of all rational no. $Q = \bigcup \{A_i : i \in \mathbb{N}\}$.

a map $f: \mathbb{N} \rightarrow A_n$.

defined by —

$$f(m) = \begin{cases} (m-1)/2n, & \text{when } m \text{ is odd.} \\ -\frac{m}{2n}, & \text{, } m \text{ is even.} \end{cases}$$

$\Rightarrow f$ is 1-1 and onto. So A_n is countable for each $n \in \mathbb{N}$.
 Since $Q = \bigcup \{A_i, i \in \mathbb{N}\}$ is the countable union of countable sets. Hence Q is a countable set.

$\mathbb{R}$ is uncountable set.

Cardinality of $\mathbb{N}$ is $\frac{\text{No of } |\mathbb{R}| = c.}{\text{transcendental no.}}$

2.7 Domain, Co-Domain and Range of a Function

If $f: A \rightarrow B$

then the set A is known as the **domain of f** and the set B is known as the **co-domain of f** . Moreover the set of all f -images of elements of A is known as the **range of f** or image set of A under f and is represented by $f(A)$.

Thus,

$$f(A) = \{f(a) \mid a \in A\}.$$

It should be observed that $f(A) \subset B$.

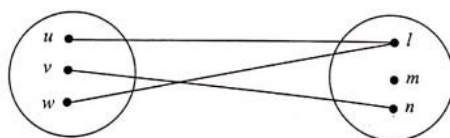
If the function is expressed as the set of ordered pairs then the domain of f will be the set of all first co-ordinates of elements of f and the range of f will be the set of the second co-ordinates of elements of f i.e.,

$$\text{domain of } f = \{x \mid (x, y) \in f\}$$

$$\text{range of } f = \{y \mid (x, y) \in f\}$$

Example 7. Let the function f be expressed as in the given diagram then

$$\text{domain of } f = \{u, v, w\}$$



$$\text{co-domain of } f = \{l, m, n\}$$

$$\text{range of } f = \{l, n\}$$

Example 12. Find the range of the functions :

$$(i) \quad y = f(x) = \frac{1}{x-5}$$

$$(ii) \quad y = x^2$$

$$(iii) \quad y = \frac{x}{1-x}$$

$$(iv) \quad y = \frac{x}{1+x^2}$$

$$(v) \quad y = \sqrt{9-x^2}$$

$$(vi) \quad y = \frac{2+x}{2-x}$$

Solution: First we should solve the given relation for x and then adopt the same criteria as we do for finding the domain

$$(i) \quad \text{From } y = \frac{1}{x-5}, x = \frac{1+5y}{y} \text{ when } y = 0, x \rightarrow \infty. \text{ So range} = R - \{0\}.$$

$$(ii) \quad y = x^2 \Rightarrow x = \pm\sqrt{y}, x \text{ is real when } y \geq 0. \text{ So range} = [0, \infty)$$

$$(iii) \quad y = \frac{x}{1-x} \Rightarrow x = \frac{y}{1+y}, x \rightarrow \infty, \text{ for } y = -1. \text{ So range} = R - \{-1\}$$

$$(iv) \quad y = \frac{x}{1+x^2} \Rightarrow x = \frac{1 \pm \sqrt{1-4y^2}}{2y}$$

$$x \text{ is real and finite when } -\frac{1}{2} \leq y \leq \frac{1}{2} \text{ and } y \neq 0. \text{ So range } f = \left[-\frac{1}{2}, \frac{1}{2}\right] - \{0\}.$$

$$(v) \quad y = \sqrt{9-x^2} \Rightarrow x = \pm\sqrt{9-y^2}$$

x is real when $0 \leq y \leq 3$, negative values are left for y to be single valued as necessary for y to be a function, therefore, range = $[0, 3]$

$$(vi) \quad y = \frac{2+x}{2-x} \Rightarrow x = \frac{2y-2}{y+1}, x \text{ is infinite when } y = -1. \text{ So range} = R - \{-1\}.$$

Handwritten calculations for Example 12:

- For (iii): $y = \frac{x}{1-x} \Rightarrow y(1-x) = x \Rightarrow y - yx = x \Rightarrow y = x + yx \Rightarrow y = x(1+y) \Rightarrow x = \frac{y}{1+y}$
- For (iv): $y = \frac{x}{1+x^2} \Rightarrow y(1+x^2) = x \Rightarrow y + yx^2 = x \Rightarrow yx^2 - x + y = 0$
- For (v): $y = \sqrt{9-x^2} \Rightarrow y^2 = 9-x^2 \Rightarrow x^2 = 9-y^2 \Rightarrow x = \pm\sqrt{9-y^2}$

Example 11. Find the domains of the following functions :

(i) $f_1(x) = \frac{x}{x^2 + 1}$

(ii) $f_2(x) = \frac{x}{x-5}$

(iii) $f_3(x) = \sqrt{x-4}$

(iv) $f_4(x) = \log x$

(v) $f_5(x) = \frac{\sqrt{x+2}}{x-2}$

(vi) $f_6(x) = \sqrt{4-x^2}$

Solution: Assuming $f(x)$ to be a real number, domain will consists of those values of x for which:

(a) $f(x)$ does not tend to ∞ ,

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(b) $f(x)$ does not become involving $\sqrt{-1}$

(c) $f(x)$ does not become indeterminate

(i) $f_1(x) = \frac{x}{x^2 + 1}$ is well defined $\forall x \in \mathbb{R}$. So $\text{dom. } f_1 = \text{set of real number } \mathbb{R}$.

(ii) $f_2(x) = \frac{x}{x-5} \rightarrow \infty$ for $x = 5$, so domain = $\mathbb{R} - \{5\}$

(iii) $f_3(x) = \sqrt{x-4}$ becomes imaginary when $x < 4$. So $\text{dom.} = \{4, \infty\}$.

(iv) $f_4(x) = \log x$, $\log x$ is real if x is positive $\log_a 0 \rightarrow \infty$, if $a > 1$ and $\log_a 0 \rightarrow -\infty$ if $a < 1$. So $\text{dom.} = \mathbb{R}^+$

(v) $f_5(x) = \frac{\sqrt{x+2}}{x-2}$ becomes imaginary for $x+2 < 0$ and tend to ∞ at $x = 2$, so domain = $(-2, \infty) - \{2\}$.

(vi) $f_6(x)$ is real if $-2 \leq x \leq 2$. So domain $f_6 = \text{interval } [-2, 2]$.

2.11 Sum and Product of Functions

Let $f: A \rightarrow \mathbb{R}$ and $g: B \rightarrow \mathbb{R}$ be two functions. Then their sum $f+g$ and product fg can be defined as below.

$$(f+g)(x) = f(x) + g(x)$$

[sum of f and g]

and

$$(fg)(x) = f(x) \cdot g(x)$$

[product of f and g]

Also

$$\text{dom}(f+g) = \text{dom}(fg) = \text{dom}(f) \cap \text{dom}(g).$$

Example 13. Let $f(x) = x^2$ and $g(x) = \sqrt{x-1}$, where $\text{dom}(f) = (-\infty, \infty)$ and $\text{dom}(g) = [1, \infty)$. Then obtain the sum and product of f and g and also find their domains.

Solution:

Sum:

$$(f+g)(x) = f(x) + g(x)$$

$$= x^2 + \sqrt{x-1}$$

Product:

$$(fg)(x) = f(x) \cdot g(x)$$

$$= x^2 \sqrt{x-1}$$

Since $\text{dom}(f) \cap \text{dom}(g) = [1, \infty)$

So both $f(x)$ and $g(x)$ are defined only when $x \geq 1$.

Thus $\text{dom}(f+g) = \text{dom}(fg) = [1, \infty)$.

Example 37. Let $f: R \rightarrow R$, defined by $f(x) = ax + b$, where $a, b, x \in R$ and $a \neq 0$. Show that f is invertible. [RTU 2010]

f is one-one : For $x_1, x_2 \in R, f(x_1) = f(x_2)$

$$\Rightarrow ax_1 + b = ax_2 + b$$

$$\Rightarrow ax_1 = ax_2$$

$$\Rightarrow x_1 = x_2 \Rightarrow f \text{ is one-one}$$

f is onto : Let $y \in R$ such that

$$y = f(x) \Rightarrow y = ax + b$$

$$\Rightarrow ax = y - b \text{ and } a \neq 0 \in R$$

$$\Rightarrow x = \frac{1}{a}(y - b) \in R$$

$\therefore$ For each $y \in R, \exists$ some $\frac{1}{a}(y - b) \in R$, such that $f(x) = y$

$\therefore f: R \rightarrow R$ is both one-one and onto.

Thus f is invertible.

Example 40. Determine whether each of the following functions is a bijection (one-to-one and onto) from R to R

(i) $f(x) = -3x + 4$

(ii) $f(x) = -3x^2 + 7$

[RTU 2011]

(iii) $f(x) = \frac{x+1}{x+2}$

(iv) $f(x) = x^5 + 1.$

[Raj. 2004]

Solution:

(i) $f(x) = -3x + 4$

One-one : Let $f(x_1) = f(x_2)$

$$\Rightarrow -3x_1 + 4 = -3x_2 + 4 \Rightarrow -3x_1 = -3x_2$$

$$\Rightarrow x_1 = x_2$$

$\therefore f$ is one-one.

Onto : Let $y = f(x) = -3x + 4 \Rightarrow x = \frac{4-y}{3}$

So for each $y \in R$ (co-domain) $\exists x = \frac{4-y}{3} \in R$ (Domain) such

$$\text{that } f(x) = -3x + 4 = -3\left(\frac{4-y}{3}\right) + 4 = y$$

Thus each element of co-domain has its pre-image in domain.
 $\therefore f$ is onto

Hence f is a bijection.

(ii) $f(x) = -3x^2 + 7$

One-one : Since $1 \neq -1$ but $f(1) = 4$ and $f(-1) = 4$

$\Rightarrow$ two elements of domain have the same f -image

$\therefore f$ is not one-one.

Hence f is not a bijection.

(iii) $f(x) = \frac{x+1}{x+2}$

Since for $x = -2$, $f(x)$ tends to ∞ , i.e., $f(x)$ is not defined for $x = -2$.

$\therefore f$ is not even a function.

Hence f cannot be a bijection.

(iv) $f(x) = x^5 + 1$

One-one : Let $f(x_1) = f(x_2)$

$$\Rightarrow x_1^5 + 1 = x_2^5 + 1 \Rightarrow x_1^5 = x_2^5 \Rightarrow x_1 = x_2$$

$\therefore f$ is one-one.

Onto : Let $y = x^5 + 1 \Rightarrow x = (y - 1)^{1/5} \in R, \forall y \in R$

So for each $y \in R$ (co-domain) $\exists x = (y - 1)^{1/5} \in R$ (domain) such that

$$f(x) = x^5 + 1 = \{(y - 1)^{1/5}\}^5 + 1 = y - 1 + 1 = y$$

Thus each element of co-domain has its pre-image in domain.

$\therefore f$ is onto.

Hence, f is a bijection.